ELSTE Data science

Class 1: Introduction to data analysis and inference using Python

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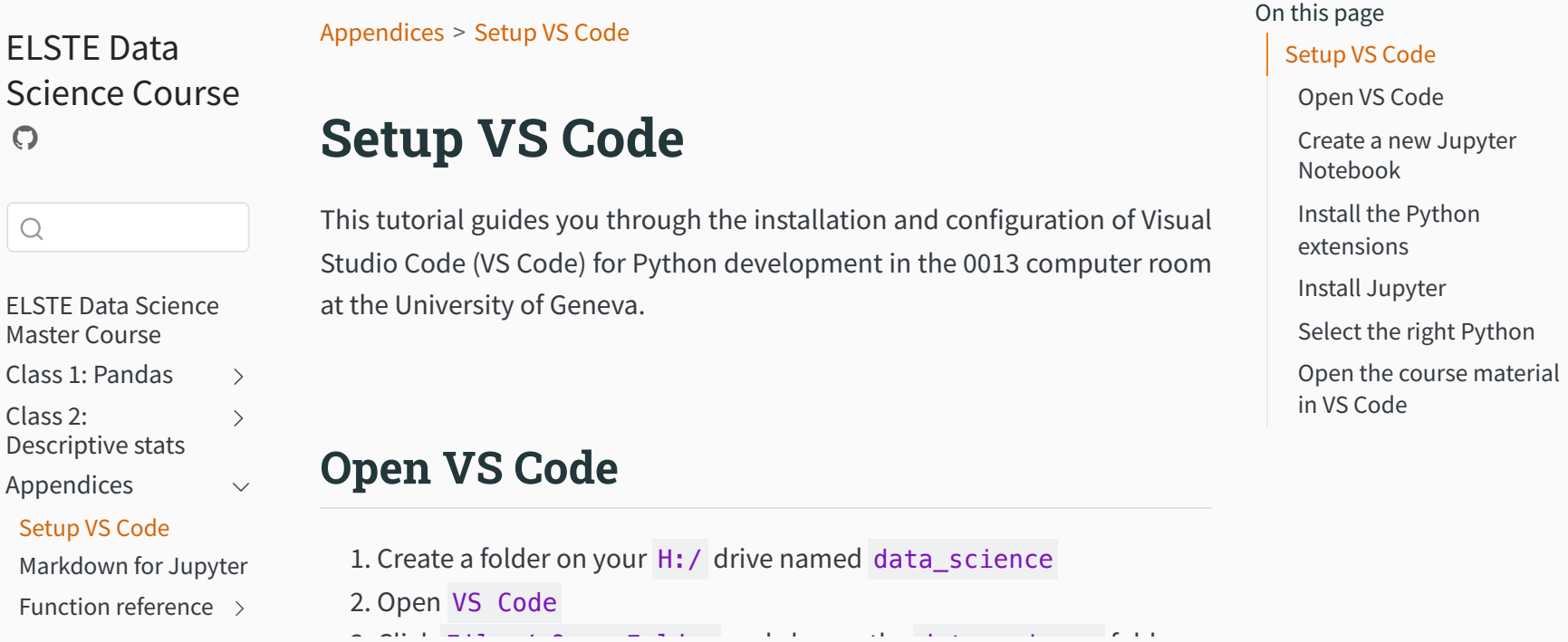
Earth Sciences

Pharmaceutical Sciences

October 14, 2026

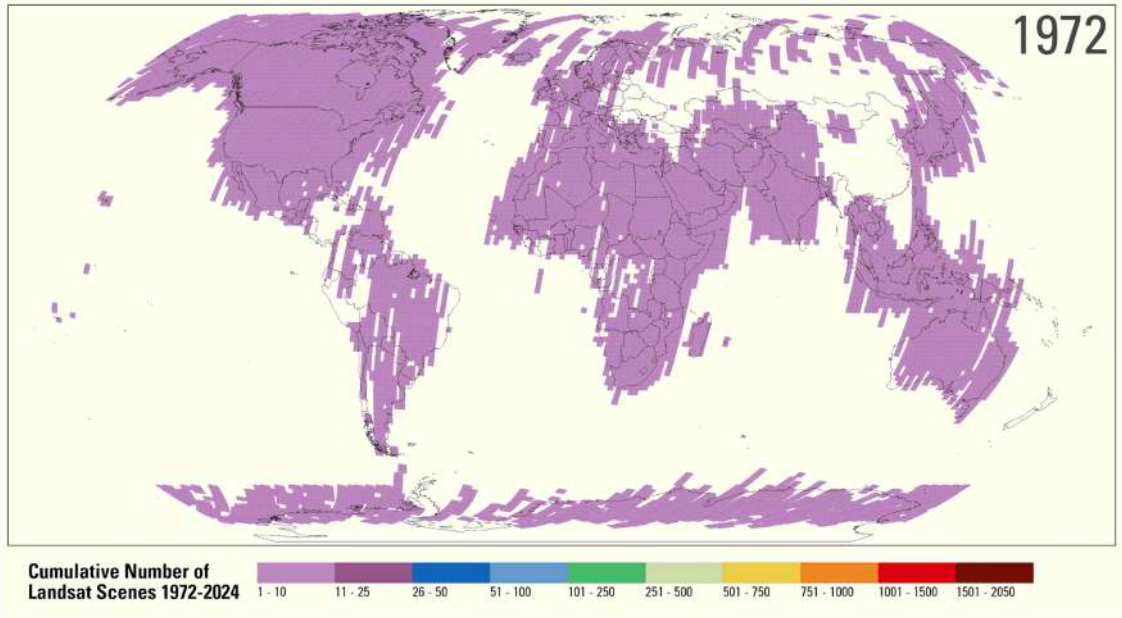
Get ready

- Go to https://elste-master.github.io/Data-Science/
  - → **Download slides:** Class 1 > Overview
  - → **Setup VS Code:** Appendices > Setup VS Code



Why this class? (1/2)

Data deluge: The big EO data landscape



- **Landsat:**
  - → Download: 53 images/day (2001) - 220’000 images/day (2017)
  - → 5 million images of the Earth surface → > 5 PB
- **ESA Sentinel 1 & 2 →** 4.6TB /daily!

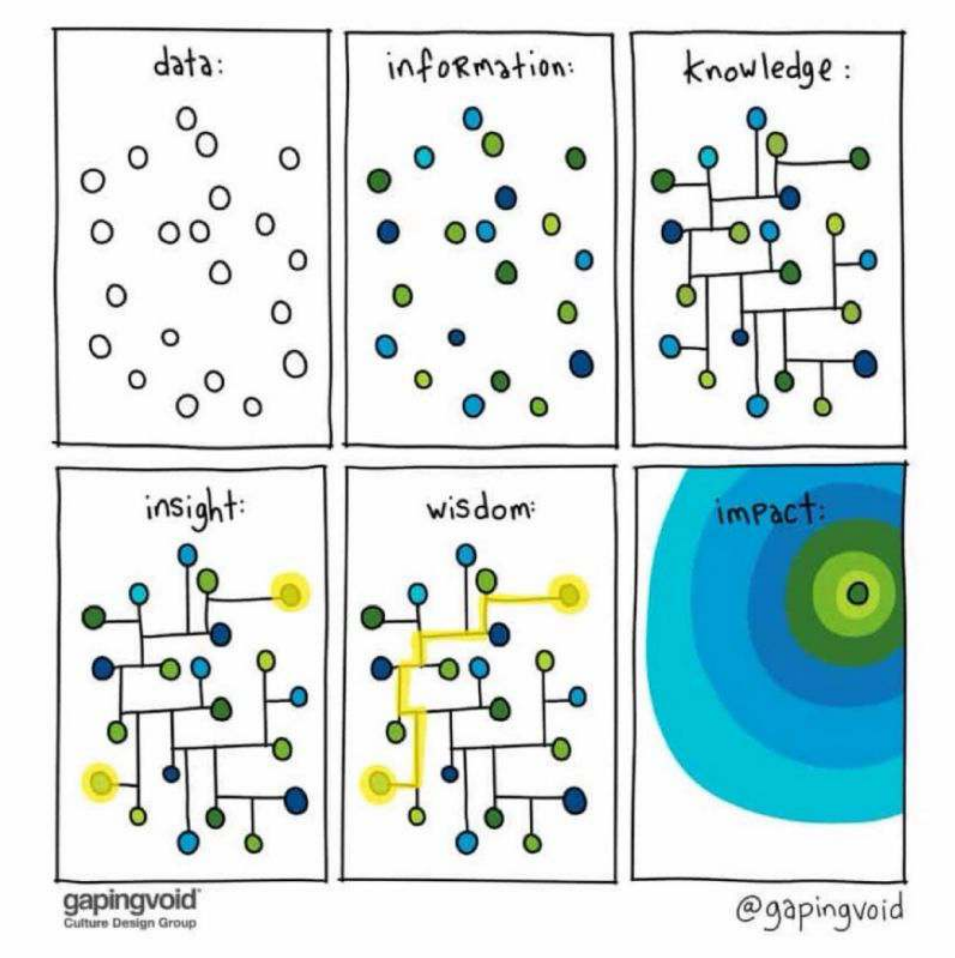
Why this class? (2/2)

As scientists, we are exposed to:

- Increasing computational power/facilities
- Increasing amount of data

How do we make sense of it all?

- Scientific coding → gateway to scientific data analysis
- Data science → extracting meaningful infomation and insights from data



What is data science

Wikipedia

> *Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge from [...] data.*

National Institutes of Standards and Technology

> *The field that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data.*

Class schedule

- **Fall part at UNIGE**: 6 × 3-h long sessions
- **Assistants**: Sofia Hernandez Castro, Henry Hoult
- **Format**: Theory, live coding and exercises

October 14 and 21

- **Seb** → Exploratory data analysis

October 28, November 4 and 11

- **Stéphane** → Inference

November 18

- Question session + exam

Class objectives

- Scientific coding and stats often appear daunting
- True, there can be a steep learning curve

Objectives

- **Not** to make you and expert developer / statistician **but**
  - → Motivate scientific reasoning
  - → Develop statistical thinking

Get involved!!!

We want this class to be useful for your research!

Try to contextualise the course material to **your research**:

- Do you already have your own datasets you could bring to class?
- Can you use the course to formulate new research questions?
- Do you know any open-access datasets relevant to your fields?

  → If not, can you find some for the next weeks?

Open data in geosciences

Database	What can you find?
PANGAEA	About **447,000 datasets**: marine geology, geochemistry, palaeoclimate and ecology. Documented datasets with citable DOIs.
EarthScope	Continuous seismic records from thousands of stations. **590 TB** archived by 2019; over **215 TB** of new geophysical data in 2024.
NOAA Paleoclimatology	Over **10,000 datasets** from ice cores, tree rings, corals and sediments, extending climate records over thousands to millions of years.
EarthChem	Geochemical databases searchable by location, age, tectonic setting, sample and composition.
IODP[3]	Ocean-drilling measurements, images and metadata, normally open after a **one-year embargo**; physical cores available from repositories.

Open Earth-observation data

Database	What can you find?
Landsat	Over **13.8 PB** of imagery since 1972. About **1 TB/day** of new observations and **3 TB/day** of derived scientific products.
Copernicus Sentinel	Radar, optical and atmospheric observations. Sentinel 1 & 2 acquire **4.6 TB/day**; the former portal distributed about **590 PB** to **750,000 users** before closing in 2023.
NASA EOSDIS	Observations of the atmosphere, oceans, ice, biosphere and land. Higher-resolution missions drive archive growth and analysis in the cloud.
ERA5	Hourly global climate reanalysis: **200+ variables** on an approximately **31 km** grid. About **5 PB** of native data and **1 PB** prepared for distribution.
NOAA NCEI	Open weather, ocean, coastal and geophysical archives from historical records, stations, buoys, ships and other observing systems.

Why coding?

Motivations (1/6)

- **Example**: A common - but unnecessarily complicated - workflow of many specialised softwares



- **Objective**: Integrate full research workflow in an environment supporting all tasks



Motivations (2/6)

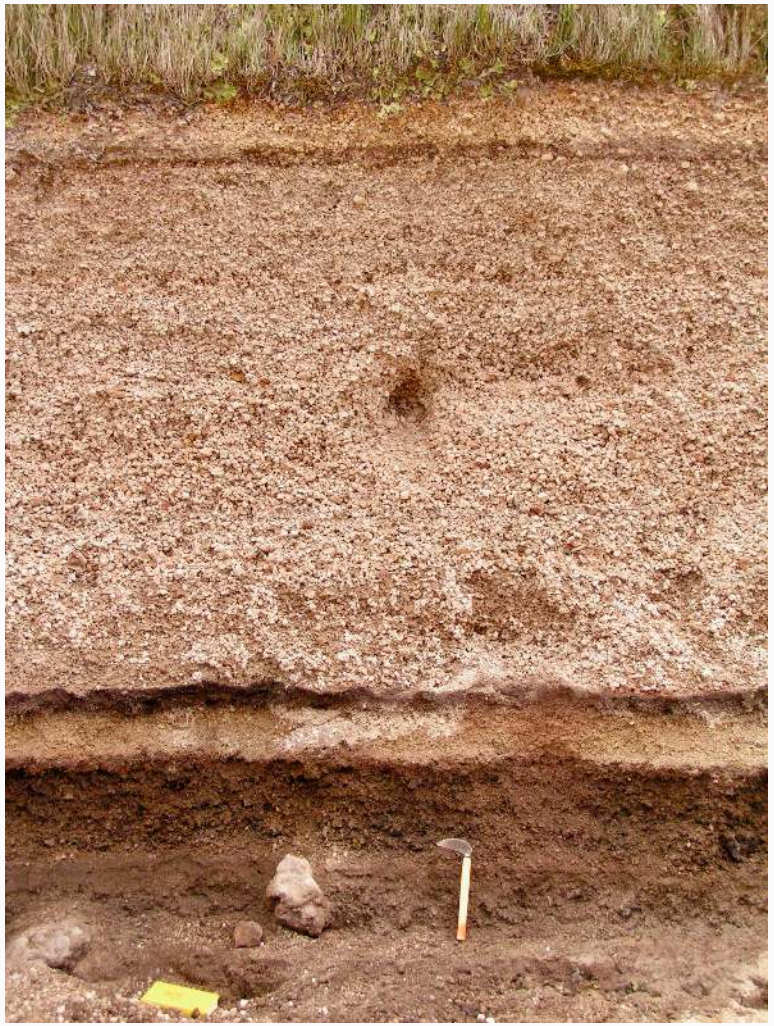
Motivation 1: Automation

Cotopaxi volcano

- Reconstructing eruption source parameters (ESP) from tephra deposits

  → *How do different measurement methods influence ESP estimates?*

- 40 outcrops
- ~10 samples per outcrop
- ~7 measurement methods per sample



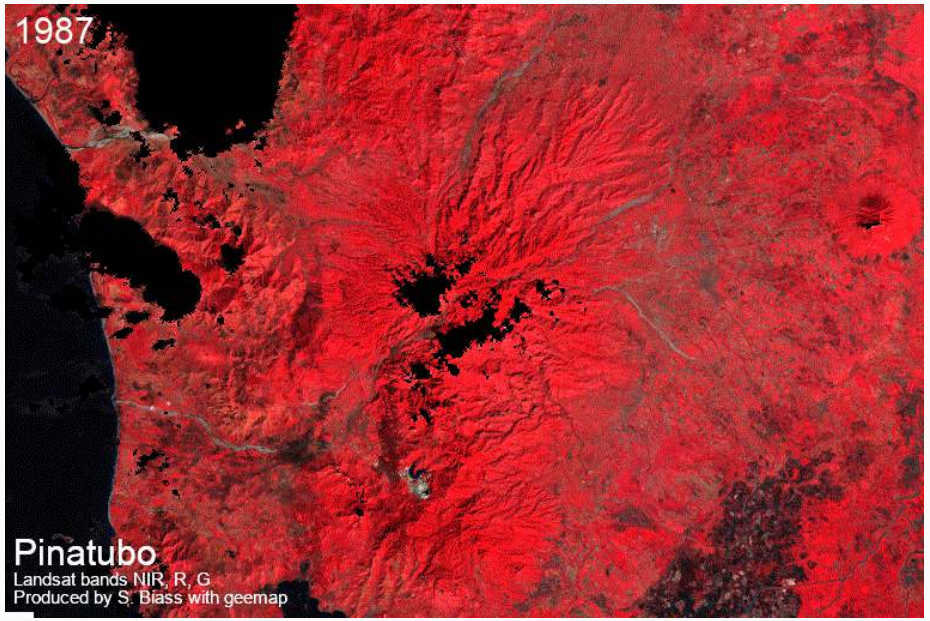
Motivations (3/6)

Motivation 2: Data analysis

Example 1: Exploit catalogues of big Earth Observation data

- **Revisit big EO catalogues to infer new knowledge**

  → *What controls the impact and recovery of vegetation following eruptions?*

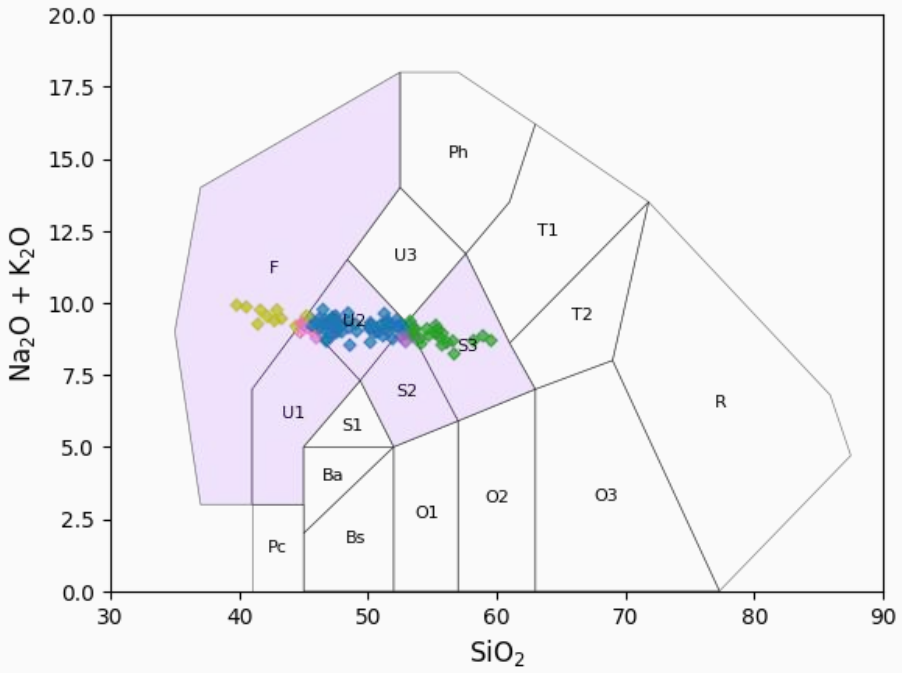


Motivations (4/6)

Motivation 2: Data analysis

Example 2: Streamline global geochemical analyses

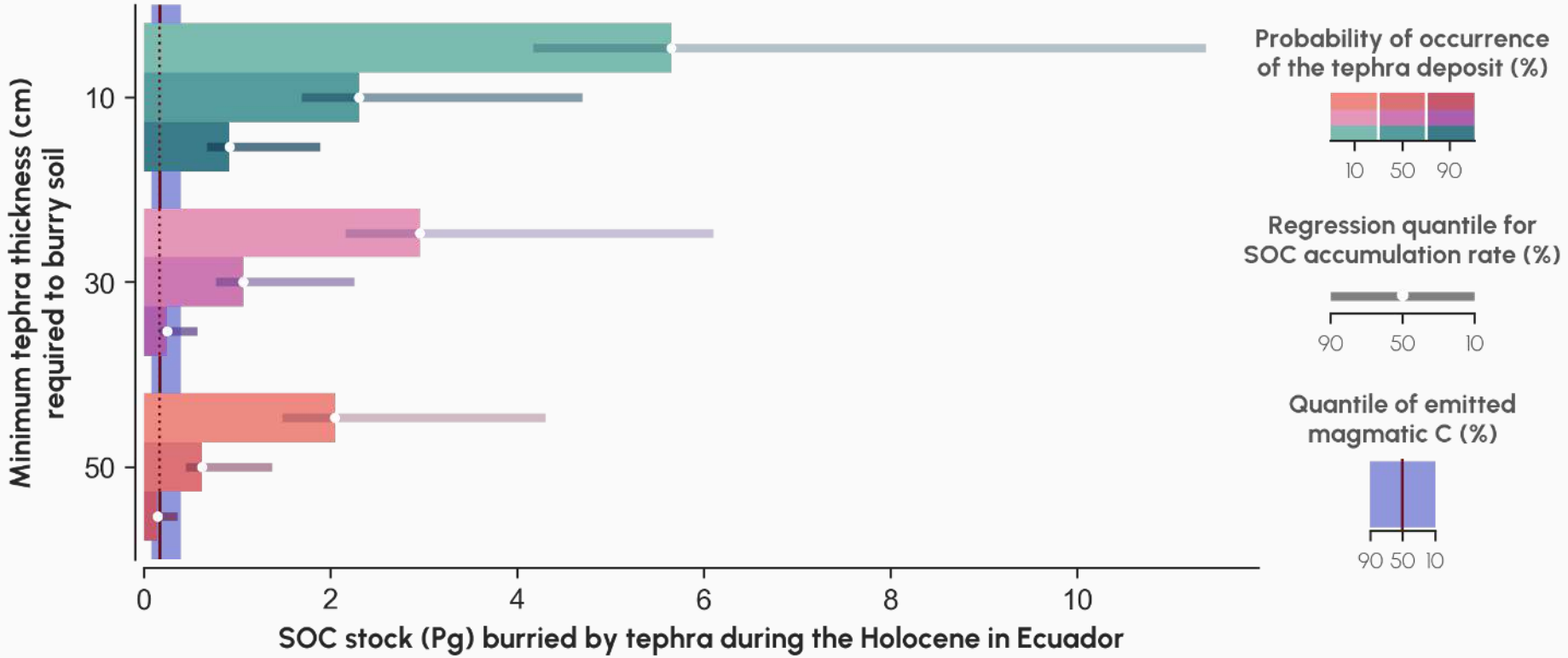
- **Access global databases**
  - → e.g., *Georock* database
- **Automatic dedicated analyses/plots**
  - → e.g., *TAS* diagrams and classification using `pyrolite`



Motivations (5/6)

Motivation 3: Visualisation

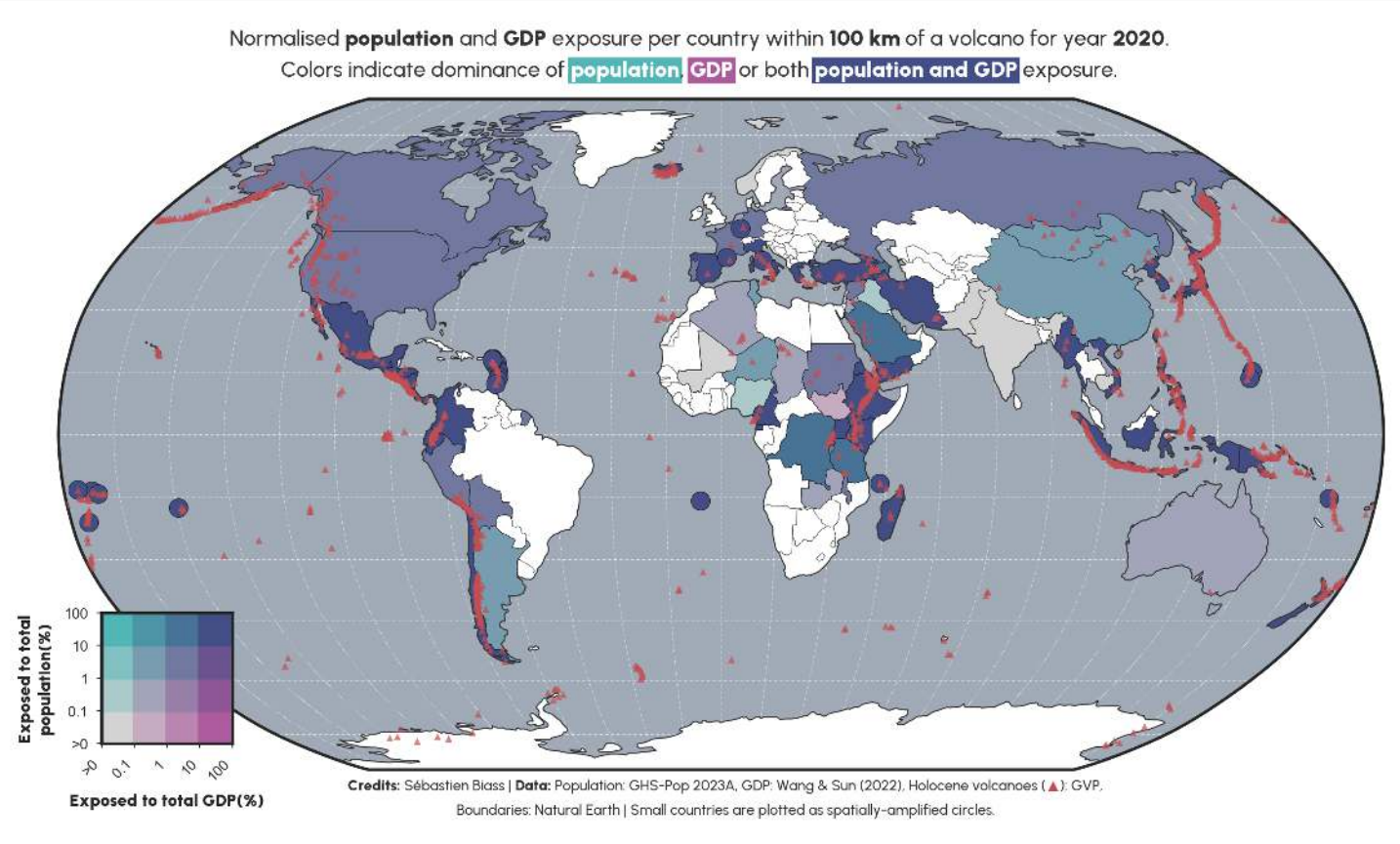
Example 1: In volcanically-active regions, soil burial after explosive eruptions capture more carbon that they emit.



Motivations (6/6)

Motivation 3: Visualisation

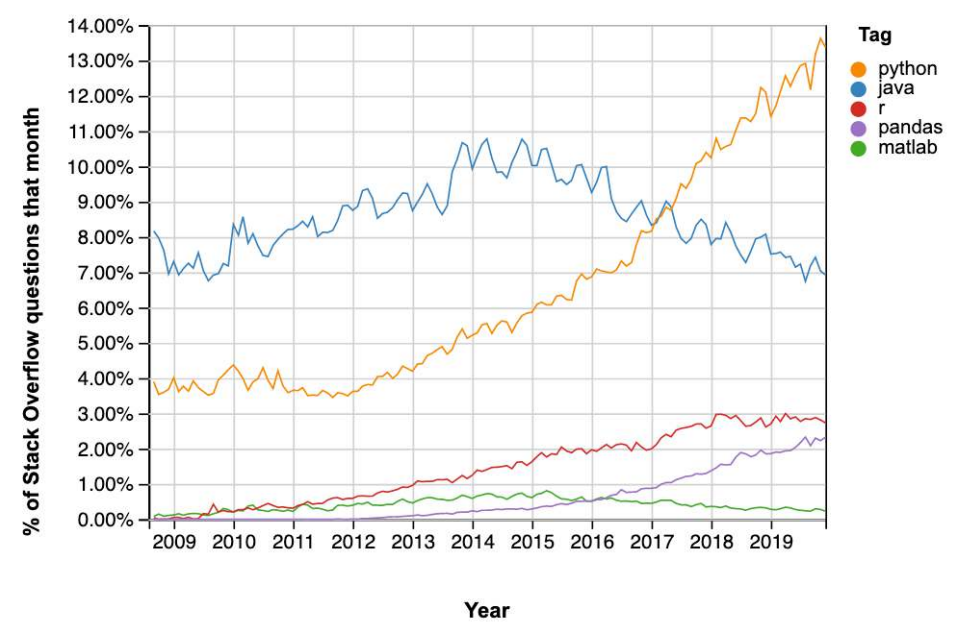
Example 2: In Small Island States, ≥90% of population and GDP is exposed to volcanic hazards.



Why Python?

Python's advantages [1]

1. **Free**
2. **Open source**
   - Not dependent on any company
   - Large online community
3. **Old** → stable (started in 1991)
4. **Very** popular in science
5. Relatively easy to learn



1. Inspired by the School of Oceanography at University of Washington

What is Python

High-level computing language

- Compiled → doesn't directly speak to the computer, but is *interpreted* by another language
  - → Usually slower, but much, much easier to use!

Modular

- Python is composed of a *core* of functions complemented by wide ecosystem of specialized packages for specific tasks
  - → A function is a bit of code to perform a **specific task**
  - → A module is a collection of **functions**
  - → A package is composed by multiple **modules**

An example of packages, modules and functions

In the code below, we first load the necessary packages and libraries:

- `matplotlib` is the main **visualisation** package used in Python
- `pyplot` is a module of `matplotlib` that provides easy-to-use functions for **plotting** data
- `figure` is a function of `pyplot` and is the main function to prepare a plot

```
1 # Import the packages and modules
2 from matplotlib import pyplot as plt
3 # Set up a figure for plotting
4 plt.figure(...)
```

How to use Python

Option 1: Your own computer

- Requires some setup
- No internet connection required
- Slow if computer is slow
- All data stored on your own computer

> Try this tutorial

Option 2: In the cloud → Google Collab

- No setup required
- No internet connection required
- Decent speed, free as long as Google says it is free
- All data stored in the cloud

How to run Python

3 main ways to run Python…

1. Run Python `.py` scripts from the command line → *deprecated*
2. Run blocks of code from within a Python `.py` script using `ipykernel` → **similar behaviour to R/Matlab**
3. Use **Jupyter Notebooks** in `.ipynb` files

…in two different environments

1. Dedicated software: VSCode, PyCharm, Spyder
2. Web browser: JupyterLab

How to approach coding

Python as a natural language

Natural language	Python	Example
Noun	Object or variable	`df`
Verb	Function or method	`mean()`, `sort_values()`
Complement	Target variable	`"altitude"`
Adverb	Argument specifying how	`ascending=False`
Sentence	Instruction	`df.sort_values(...)`

```
1 df.sort_values("altitude", ascending=False)
```



Python checks the grammar of your code. **You** must check the scientific meaning of the sentence.

From a sentence to a sequence of tasks

Task	Questions to ask
Load	Where are the data? How do I read them?
Inspect	What do the rows, columns and values represent?
Select / filter	Which columns and observations do I need?
Transform	What should I sort, recode or calculate?
Summarise / analyse	What quantity or relationship answers my question?
Visualise / export	How should I show or save the result?

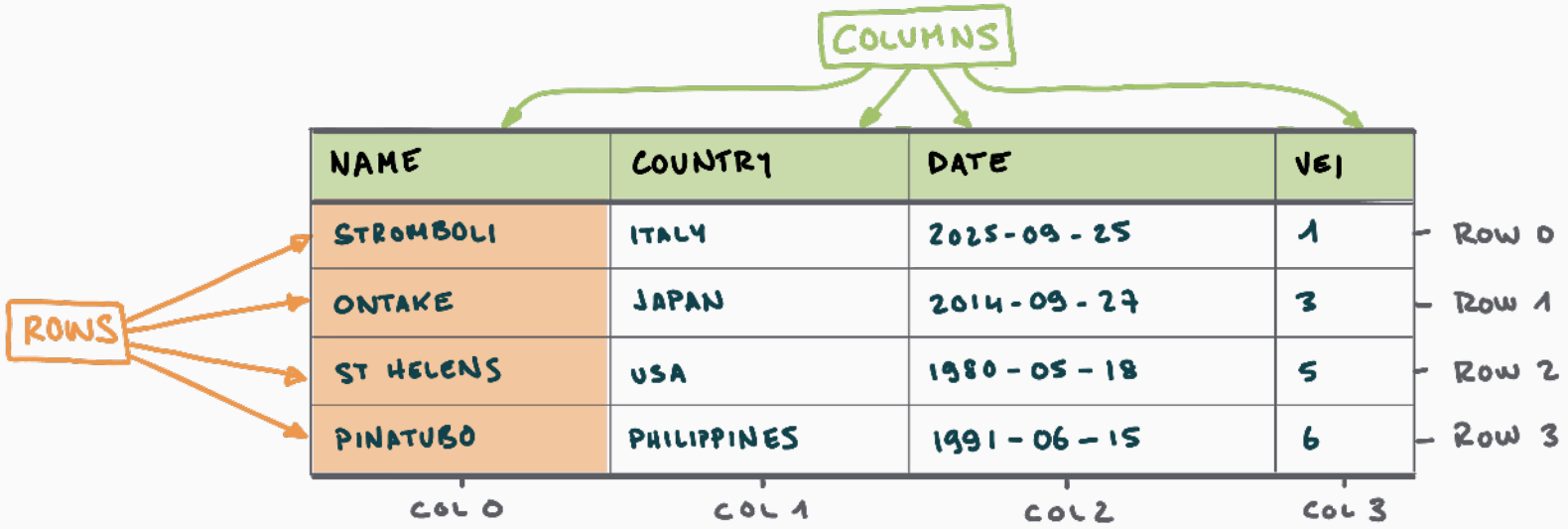
Today’s plans

Pandas

`pandas`: A **package** for data manipulation and analysis handling **structured data**

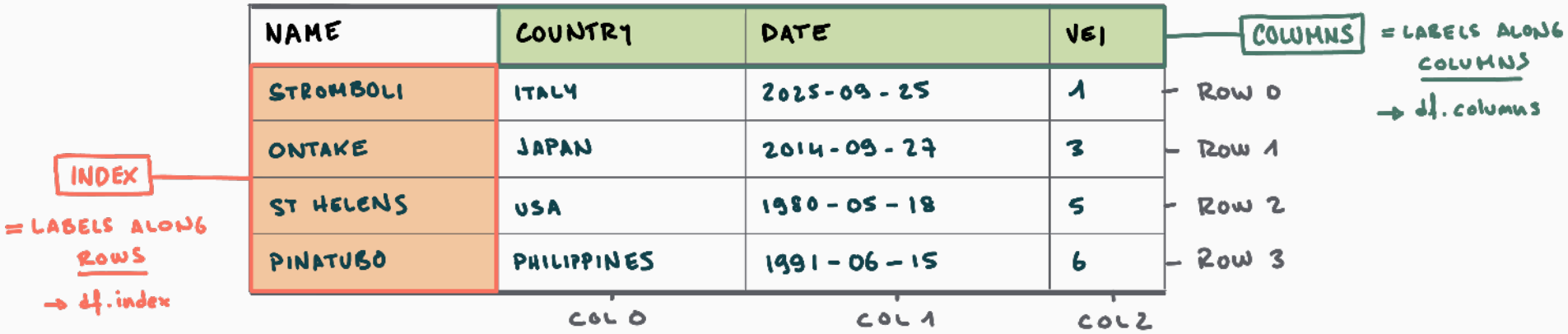
- **Reading/writing data** from common formats (CSV, Excel, JSON, etc.)
- Handling **missing data**
- **Filtering**, **sorting**, **reshaping** and **grouping** data
- **Aggregating** data (sum, mean, count, etc.)
- **Time series support** (date ranges, frequency conversions)
- **Statistical operations**

Anatomy of a DataFrame (1/2)



- Similar to *Excel* → contains *tabular data* composed of rows and columns
- In *Excel*:
  - → **Rows** are accessed using *numbers*
  - → **Columns** are accessed using *letters*

Anatomy of a DataFrame (2/2)



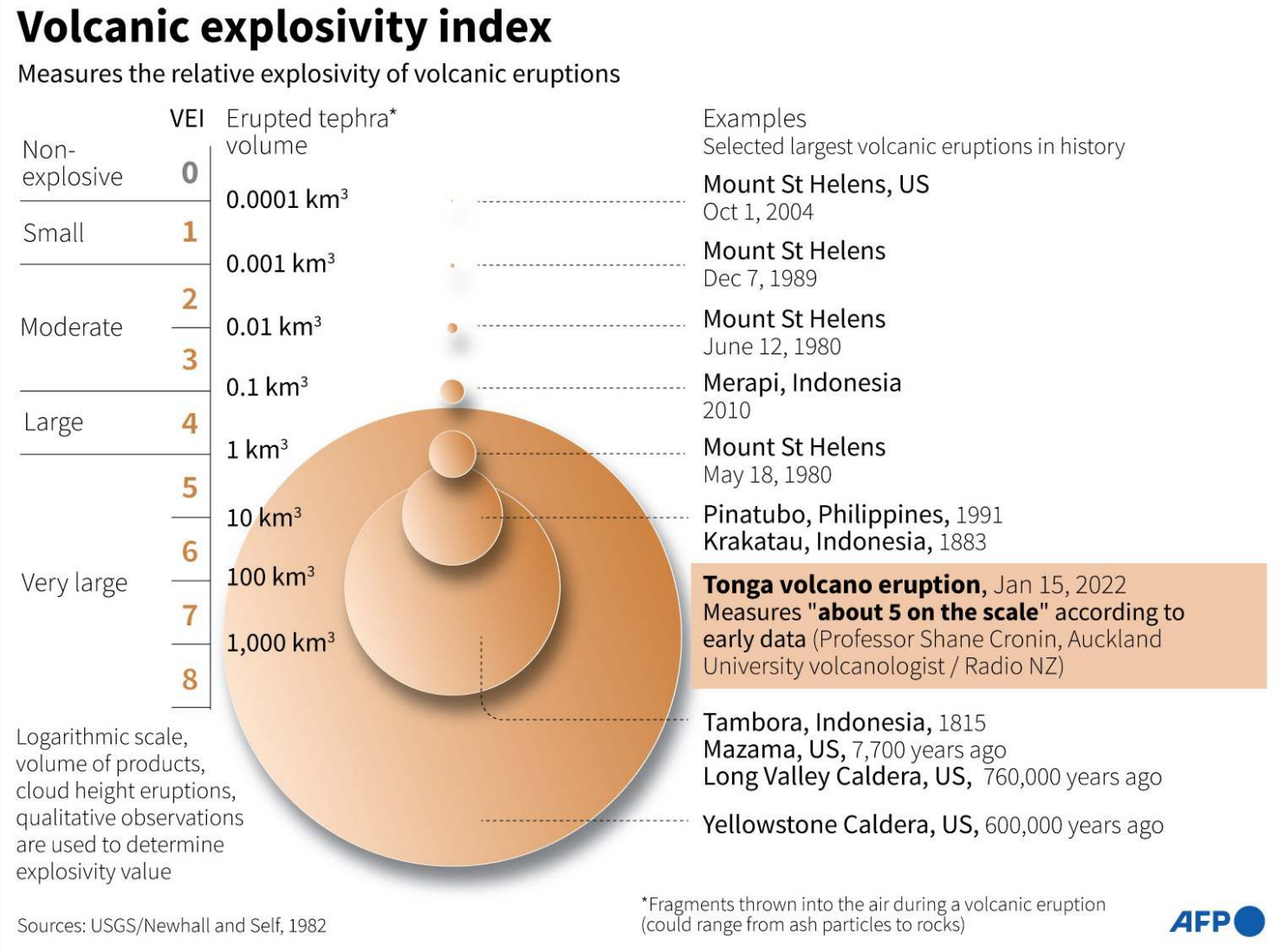
- Unlike *Excel*, **rows** and **columns** can be labelled
  - → Index refers to the label of the **rows**. In the *index*, **values are usually unique** - meaning that each entry has a different label.
  - → Column refers to the label of - logically - the **columns**

Today's dataset

Synthetic dataset of **selected volcanic eruptions** → first 5 rows:

Name	Country	Date	VEI	Latitude	Longitude
St. Helens	USA	1980-05-18	5	46.1914	-122.196
Pinatubo	Philippines	1991-04-02	6	15.1501	120.347
El Chichón	Mexico	1982-03-28	5	17.3559	-93.2233
Galunggung	Indonesia	1982-04-05	4	-7.2567	108.077
Nevado del Ruiz	Colombia	1985-11-13	3	4.895	-75.322

Volcanic explosivity index (VEI)



Today's objectives

5 modules:

1. How to load data in a DataFrame
2. How to access data → *query by label/position*
3. How to filter data → *comparison and logical operators*
4. How to rearrange data → *sorting values*
5. How to operate on data → *arithmetic and string operations*

- We expect you to know how to get started with Python / Jupyter notebook
- Let us know if you are lost or if you already know all of this

  → Let's make it useful for **your research**

Get started

- Go to https://elste-master.github.io/Data-Science/
  - → **Setup VS Code:** Appendices > Setup VS Code
  - → Follow Class 1

How to do it?

- Create a Jupyter Notebook
- Copy & paste code fragment → make sure you **understand** what you are doing!
  - → Use the documentation / **cheat sheets** / **companion course**
- Document your code with **Markdown**